

Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Code Layout, Readability and Reusability

This document evaluates the quality of the Emotion Detection & Learning Support Engine source code. The application follows a modular architecture where authentication, emotion detection, AI guidance, analytics, database operations, and utility functions are separated into dedicated modules. Proper naming conventions, reusable components, exception handling, and documentation improve maintainability and future scalability.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code.	Yes	Adheres strictly to standard PEP 8 spacing formatting across all Python templates.
2	Proper File Structure	Files and folders are logically organize.	Yes	Modular project architecture
3	Meaningful Variable Names	Variables reflect their purpose clearly.	Yes	Easy to understand
4	Function / Method Names	Functions are descriptively named	Yes	Self-explanatory methods
5	Code Comments	Inline and block comments explain logic.	Yes	Major modules documented
6	Modular Design	Code is split into reusable functions.	Yes	Authentication, AI, Database, Analytics separated
7	No Redundant Code	Duplicate or unused code is removed	Yes	Utility functions reused
8	Error Handling	Exceptions and errors are handled carefully.	Yes	API, Database and File exceptions managed

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	Emotion Classifier	Python	Emotion Prediction Pipeline	High

2	Gemini Service	Google Gemini API	AI Learning Guidance	High
3	Database Manager	SQLite	Store & Retrieve Emotion Records	High
4	Analytics Dashboard	Streamlit + Plotly	Analytics Visualization	Medium
5	Text Utilities	Python	Text Cleaning & Preprocessing	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure		
Readability		
Reusability		
Documentation / Comments		
Overall Score		